CPE 464 Lab 1 **Worksheet**

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This lab worksheet needs to be typed. Each student must type in their own answers. You will upload your worksheet to Canvas. This worksheet is due on the Friday after lab at 11:59 pm.

In this lab we are going to look at some basic networking concepts and tools. The steps in this lab are:

A. Program #1 questions
B. Ping
C. Traceroute (tracert on windows)
D. Wireshark for packet analysis

This lab assumes you have already installed Wireshark. (https://www.wireshark.org/#download)

## 1) Ping

The ping command allows you to query another computer to see if that computer is correctly connected to a network.

The format for the ping command is: **ping <*hostname*>**
You can execute this command in a Linux terminal or in the windows command window. On a Linux/Mac us: ping -c 5 <hostname>. This will limit the ping command to 5 requests/replies.

a. ping a University. e.g. ping www.msu.edu. (DO NOT use www.msu.edu... Pick some other university in the US and try them. If the entire class goes to the same university, they may lock us out. Keep trying until you get all successful pings for a site.)

   What is the average time it takes for this ping to be successful? (note – a ping is measured by sending and then receiving a ping packet – so it's a round trip. We can refer to this time as RTT or round trip time.)

```
(venv) rccramer@rccramer:~$ ping -c 5 www.ucsb.edu
PING www.ucsb.edu (2620:12a:8000::3) 56 data bytes
64 bytes from 2620:12a:8000::3: icmp_seq=1 ttl=53 time=12.7 ms
64 bytes from 2620:12a:8000::3: icmp_seq=2 ttl=53 time=15.8 ms
64 bytes from 2620:12a:8000::3: icmp_seq=3 ttl=53 time=15.5 ms
64 bytes from 2620:12a:8000::3: icmp_seq=4 ttl=53 time=17.1 ms
64 bytes from 2620:12a:8000::3: icmp_seq=5 ttl=53 time=15.7 ms

--- www.ucsb.edu ping statistics ---
5 packets transmitted, 5 received, 0% packet loss, time 4006ms
rtt min/avg/max/mdev = 12.743/15.382/17.147/1.441 ms
(venv) rccramer@rccramer:~$
```

**The average RTT from my computer to www.ucsb.edu is 15.382 milliseconds.**

b. ping www.imperial.ac.uk. What is the average time (RTT) it takes for this ping to be successful?

```
(venv) rccramer@rccramer:~$ ping -c 5 www.imperial.ac.uk
PING www.imperial.ac.uk (2a0c:5bc0:88:100:1::172) 56 data bytes
64 bytes from wrpwww.cc.gslb21.ic.ac.uk (2a0c:5bc0:88:100:1::172): icmp_seq=1 ttl=40 time=195 ms
64 bytes from wrpwww.cc.gslb21.ic.ac.uk (2a0c:5bc0:88:100:1::172): icmp_seq=2 ttl=40 time=204 ms
64 bytes from wrpwww.cc.gslb21.ic.ac.uk (2a0c:5bc0:88:100:1::172): icmp_seq=3 ttl=40 time=228 ms
64 bytes from wrpwww.cc.gslb21.ic.ac.uk (2a0c:5bc0:88:100:1::172): icmp_seq=4 ttl=40 time=249 ms
64 bytes from wrpwww.cc.gslb21.ic.ac.uk (2a0c:5bc0:88:100:1::172): icmp_seq=5 ttl=40 time=171 ms

--- www.imperial.ac.uk ping statistics ---
5 packets transmitted, 5 received, 0% packet loss, time 4428ms
rtt min/avg/max/mdev = 170.554/209.311/249.385/27.098 ms
(venv) rccramer@rccramer:~$
```

**The average RTT to www.imperial.ac.uk is 209.311 milliseconds**

c. We talked about latency (RTT or round trip time) in lecture. Can you speculate why it takes longer to ping www.imperial.ac.uk than the US based university?

It likely takes www.imperial.ac.uk longer than www.ucsb.edu because of the physical distance, which means an increased number of hops. This means an increased transmission delay, because we have to handshake with more routers along the way than ucsb.edu.

We will look at the ping command again as part of our use of wireshark later in this lab.

2) **Traceroute** (on windows its **tracert** or **tracepath** on newer Linux boxes)

The traceroute (tracert, tracepath) command allows you to see the path taken across the Internet. It returns the IP address or name of the routers that the packet traverses. Let's look at the path between your location and some other locations.[1]

a. In a terminal/command window type: traceroute www.imperial.ac.uk (or on windows: tracert www.imperial.ac.uk or on some Linux boxes: tracepath www.imperial.ac.uk)

i. How many hops (routers) are between you and www.imperial.ac.uk?

```
(venv) rccramer@rccramer:~$ tracepath www.imperial.ac.uk
 1?: [LOCALHOST]                      0.015ms pmtu 1500
 1:  syn-2600-6c52-7c00-009c-0000-0000-0000-0001.biz6.spectrum.com   2.840ms
 1:  syn-2600-6c52-7c00-009c-0000-0000-0000-0001.biz6.spectrum.com   2.227ms
 2:  no reply
 3:  lag-58.hcr01snloca88.netops.charter.com              39.440ms asymm  4
 4:  no reply
 5:  lag-9.crr01snloca.netops.charter.com                 14.806ms asymm  6
 6:  no reply
 7:  lag-800.bbr02snloca.netops.charter.com               19.079ms asymm  6
 8:  lag-805.bbr01snjsca.netops.charter.com               20.083ms asymm  9
 9:  lag-803.prr02snjsca.netops.charter.com               15.897ms asymm 10
10:  no reply
11:  system.0.ear3.London2.level3.net                    184.551ms asymm 17
12:  2001:1900:5:2:2:0:110:1e2                           204.445ms asymm 24
13:  ae24.londtt-sbr1.ja.net                             203.341ms asymm 19
14:  ae28.londtw-sbr2.ja.net                             306.347ms asymm 18
15:  ae26.londtw-ban1.ja.net                             233.738ms asymm 22
16:  ic.londtw-ban1.ja.net                               204.049ms asymm 20
17:  no reply
18:  no reply
19:  no reply
```

**There are 16 known hops between myself and www.imperial.ac.uk. I can tell because after we reach the destination, it stopped replying.**

ii. Speculate which hop takes you from the US to England, which hop is it?

**It appears that the hop that takes me from the US to England could be lag-803.prr02snjsca.netops.charter.com because after that, the next hop has London in the name.**

[1] Because we are using live networks with real sites, sometimes those sites are not available or think Cal Poly is sending them too much weird traffic and stop responding. If this happens just pick another university (or really any site).

*iii.* About the first router in England:

From the traceroute, what is the name of the first router you think is in England:

Let's look up information on that router by its name. Go to Google and search for the name of the router. (For me the router is *JANET.ear3.London2.Level3.net*, so I typed '<u>who has JANET.ear3.London2.Level3.net</u>' in Google. Note, you could also look it up by IP address.)

In what country (and city) is this router located?

**The router's name is ldn-bb1-v6.ip.twelve99.net and it is located in London, UK.**

b. Traceroute to www.umich.edu (traceroute www.umich.edu)

i. How may hops are between you and www.umich.edu?

**There appears to be 10 hops between me and www.umich.edu.**

```
(venv) rccramer@rccramer:~$ tracepath www.umich.edu
 1?: [LOCALHOST]                      0.017ms pmtu 1500
 1:  syn-2600-6c52-7c00-009c-0000-0000-0000-0001.biz6.spectrum.com   3.527ms
 1:  syn-2600-6c52-7c00-009c-0000-0000-0000-0001.biz6.spectrum.com   1.694ms
 2:  no reply
 3:  lag-58.hcr01snloca88.netops.charter.com              20.662ms asymm  4
 4:  no reply
 5:  no reply
 6:  lag-300.bbr01snloca.netops.charter.com               17.374ms asymm  7
 7:  lag-5.bbr01mtpkca.netops.charter.com                 19.066ms asymm  8
 8:  lag-802.prr01lsanca.netops.charter.com               17.983ms asymm  9
 9:  2400:cb00:12:200::1e                                 20.060ms asymm 10
10:  2400:cb00:1023:3::                                   54.865ms asymm 12
11:  no reply
12:  no reply
13:  no reply
14:  no reply
15:  no reply
16:  no reply
17:  no reply
```

ii. How many routers are the same in the traceroute results for www.imperial.ac.uk and www.umich.edu?

**There are two shared ones, syn-2600-6c52-7c00-009c-0000-0000-0000-0001.biz6.spectrum.com and lag-58.hcr01snloca88.netops.charter.com.**

3) **Wireshark**.

In this part of the lab you need Wireshark (www.wireshark.org) on your computer. Wireshark allows you to analyze packet traces. These packets may be captured on a live network (this is called packet sniffing) or wireshark can analyze packet traces previously captures and stored in a file with the extension '.pcap' or '.pcapng.' In this lab (and in program 1) you will analyze packet trace files that I captured on my network.

Note…Your first program, called trace, is to write a simple version of a packet analyzer. You will be given packet trace files (.pcap) and are required to output information on each packet.

a. Download and install wireshark (www.wireshark.org) on your computer
b. Download the ping.pcap file from canvas
c. Open the ping.pcap file in wireshark

First, you are going to look at to look at the headers for ping packets. For the first part of the lab, there are three headers that we are interested in. These are the Ethernet header, IP header and ICMP (ping) header.

```
> Ethernet II, Src: Agere_90:75:89 (00:02:2d:90:75:89), Dst: LinksysG_78:c4:7d (00:06:25:78:c4:7d)
> Internet Protocol Version 4, Src: 192.168.1.102, Dst: 64.233.187.99
> Internet Control Message Protocol
```

Using Wireshark, you will expand each of these headers and look at the values in these headers.

## 4) Ethernet Overview (using the ping.pcap file)

a. Based on the lecture, draw (or list the fields in) a standard ethernet header, include the length of each field. (If you don't remember google ethernet header).

The fields are **| 6 bytes dest MAC | 6 bytes src MAC | 2 bytes type |**

Looking at the Ethernet header in 4th packet in the ping.pcap file via wireshark. Expand the Ethernet PDU so you can see all of the fields.

b. What is the source Ethernet MAC address in the colon notation (e.g. 05:03:…)?

**The source MAC address is 00:06:25:78:c4:7d**

c. What is the destination Ethernet MAC address?

**The destination MAC address is 00:02:2d:90:75:89**

d. This Ethernet frame caries an IP payload, what is the Ethernet **Type** value that specifies the this?

**The Ethernet Type value is 0x0800 (IPv4).**

What is the hex value for the **type** field for the following payloads that may be carried by Ethernet (see https://en.wikipedia.org/wiki/EtherType):

**e.** **Type** field value for Internet Protocol version 4 (IPv4)? **0x0800**
**f.** **Type** field value for Internet Protocol version 6 (IPv6)? **0x86DD**
g. **Type** field value for Address Resolution Protocol (ARP)? **0x0806**

h. Explain how you could determine that an Ethernet Frame carries a TCP segment (talk about it with your group)?

**If you want to determine that an Ethernet Frame carries a TCP segment, get the packet, then examine the type. If the type is IPv4 or IPv6, inspect the IP header. This will tell you whether or not the frame carries a TCP packet.**

## 5) Internet Protocol Overview (using the ping.pcap file)

Go to the 1st packet in the file. The ping protocol is called ICMP. Expand the packet in the middle window (right click then "Expand All").

a. What is the **source** IP address for the 1st ping request

**The source IP address is 192.168.1.102.**

b. What is the **destination** IP address for the 1st ping request packet?

**The source IP address is 66.94.230.35**

Compare the packet in the ping.pcap file and the IP header figure from Wikipedia:

https://en.wikipedia.org/wiki/IPv4#/media/File:IPv4_Packet-en.svg

c. What Version of IP is this in binary and decimal[2]?
**In binary it's 0100, in decimal it's 4.**

d. What is the header length (IHL) value in binary and decimal?
**The header length is 0101 or 5, which is 20 bytes.**

e. What is the total length value?
**The total length value is 60.**

f. What is the Time to Live (TTL) value?
**The TTL value is 128.**

g. What is the Protocol value?
**The protocol value is 1 (ICMP).**

Go to Wikipedia for a list of protocol numbers:
https://en.wikipedia.org/wiki/List_of_IP_protocol_numbers

h. What is the protocol number for TCP?
**The protocol # for TCP is 6.**

i. What is the protocol number for UDP?
**The protocol # for UDP is 17.**

j. What is the protocol number for ICMP?
**The protocol # for ICMP is 1.**

6) **Ping** – officially called Internet Control Message Protocol (ICMP)

Looking at the "Internet Control Message Protocol" section of packet #1.

a. What is the ICMP **type** value for the ICMP Echo request packet (packet 1)?
**The ICMP type value for the echo request packet is 8.**

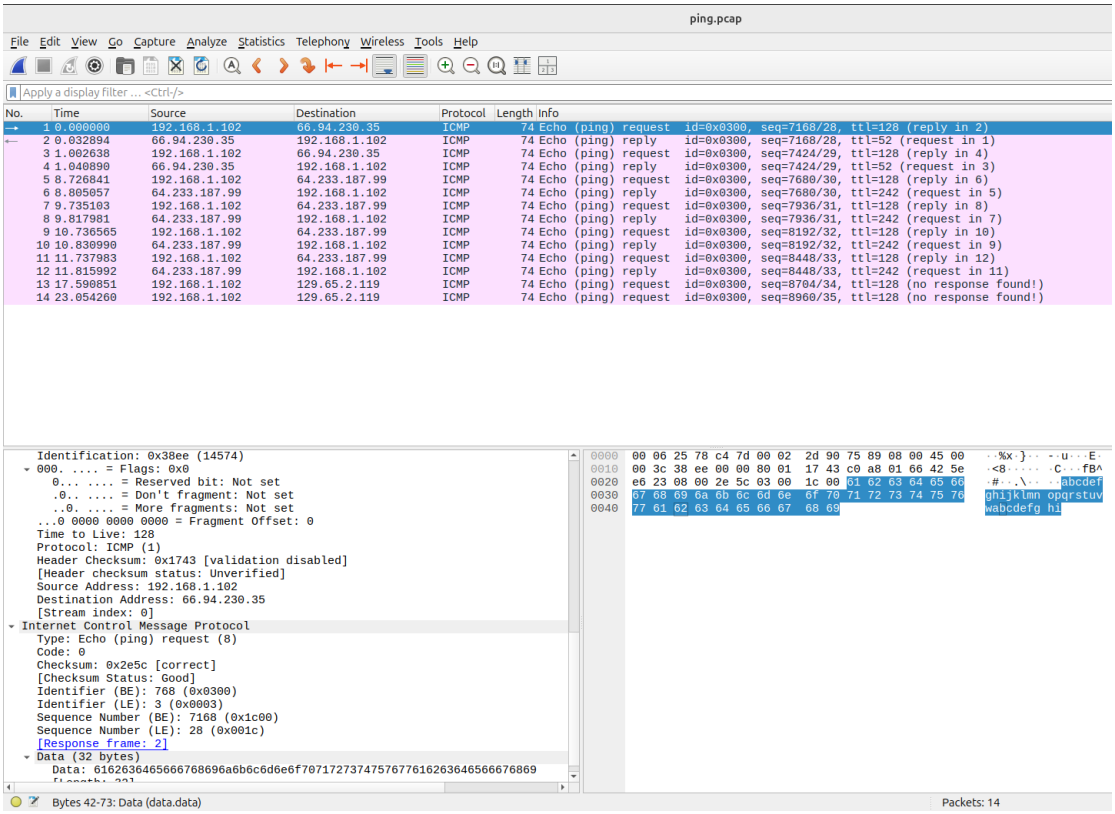
b. Look at packet 2, what is the **type** value for the ICMP Echo reply packet?
**The ICMP type value for the ICMP Echo reply packet of packet 2 is 0.**

c. Back to packet 1, what is the data that was sent in the ping request packets (in ASCII – select the data in the middle window and look at what is highlighted in the bottom window)?
**The data sent in the ping request packets was <u>abcdefghiklmnopqrstuvwabcdefghi</u> .**

d. When you turn in this lab, include a screenshot of wireshark running on your computer with the ping.pcap packets.

**Screenshot included below (on the top of the next page)**

[2] The IP version is in the first 4 bits of the first byte of the IP Header. The IHL field is the last four bits of the first byte of the IP header.




## 7) DHCP (still using wireshark)

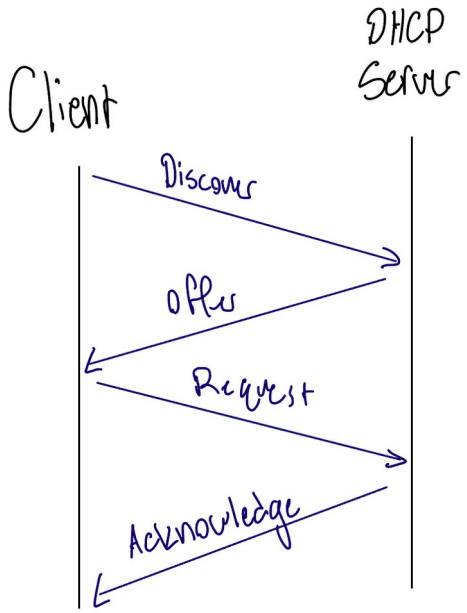
Now we are going to study one of the network protocols we use every day. This protocol is called DHCP. Using Google answer the questions below.

a) What does DHCP stand for?
**Dynamic Host Configuration Protocol.**

b) What is the purpose of DHCP?
**It's used to automatically assign IP addresses. This way network admins don't need to manually assign IP addresses to every device.**

c) Name and explain the four basic packets used in DHCP
**DORA: Discover, Offer, Request, Acknowledge**
**Discover – Broadcast packet sent by client to find available DHCP servers on network.**
**Offer – Server responds with available IP address, subnet mask, and gateway.**
**Request – Client broadcast accepts transmission from server**
**Acknowledge – Server confirms assignment, completing process.**

d) Draw a packet flow diagram of the packets that flow between your computer and the DHCP server.




e) Looking at the DHCP packet trace file (found on canvas, dhcp_trace.pcapng) using wireshark, answer the following questions:

i. Looking at the first packet, expand the Ethernet II header:

a. what is the source (src) MAC address?
**The src MAC address is 98:5f:d3:4e:bc:13**
b. In the same packet, what is the destination (dst) MAC Address?
**The dst MAC address is ff:ff:ff:ff:ff:ff**
c. The destination MAC address is a broadcast address, explain why.
**The destination MAC address is a broadcast address because for the first packet, the client doesn't know the DHCP server MAC address or IP, so it broadcasts the frame to everyine on the LAN.**

ii. Looking at the DHCP Offer packet (expand this header) answer the following questions:

a. What is the IP address of the DHCP server?
**The IP address of the DHCP server is the src or 192.168.86.1 (src since it is the offer).**
b. How long is the lease for this DHCP offer?
**The IP address lease time is 1 day (86400 seconds)**
c. What is the client **offered** (your) IP Address?
**The client offered IP address is 192.168.86.221.**
d. What is the Router's IP address (this is also called a default gateway)?
**The Router's IP address is 192.168.86.1**

f) Describe two different types of DHCP Vulnerabilities. (google DHCP security issues)

**1. DHCP servers can be spoofed, because an attacker can introduce an unauthorized DHCP server which can give clients fake IP addresses, default gateways, and DNS servers. This allows traffic interception by the spoofer.**

**2. DHCP can easily have a starvation attack where the attacker makes so many DHCP requests that the available IP pool is depleted. This created DoS (Denial of Service).**

**8) Question Deleted.**

**When you are done, I recommend you begin working on program #1 called trace.**